

Near-linear runs of strict log-concavity failures at one fixed maximum degree

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Abstract

We replace the initial edge in a recursive 64-branching tree by an even path. This decouples the unstable mean multiplier from the branching number: the former approaches 64 while the maximum degree stays at 65. For every $\varepsilon > 0$ the resulting trees have consecutive runs of strict log-concavity failures of length $\Omega_\varepsilon(N^{1-\varepsilon})$. For each fixed path, these runs remain a positive proportional distance below the independence number. We prove the required uniform local limit theorem, including its second derivatives, using an explicit small-frequency annulus and a separate finite-height lattice estimate. For the four-vertex path we additionally give a numerical certificate: every height $h \geq 512$ has a run of at least $2^{-209} N^{9/10}$ failures below 0.997α . Its verifier uses validated interval arithmetic.

Scope. The general fixed-degree result below has an analytic proof. Its constants c_s, H_s are not numerically tabulated for every path length. The separate four-vertex-path result specifies and verifies its interval, height, curvature and all numerical error bounds. The certificate is a computer-assisted component of a written proof, not a proof-assistant formalization. No claim of established publication priority or independent refereeing is made.

1 Statement and construction

For a tree T , write $i_k(T)$ for its number of independent sets of size k and $\alpha(T)$ for its independence number. A failure at k means

$$i_k(T)^2 < i_{k-1}(T)i_{k+1}(T).$$

Let $M_{65}(N)$ be the largest number of consecutive failure indices in a tree with at most N vertices and maximum degree at most 65.

For an integer $s \geq 1$, let $T_{s,0}$ be the path on $2s$ vertices, rooted at an endpoint. Form $T_{s,h+1}$ by attaching the roots of 64 disjoint copies of $T_{s,h}$ to a new root. Directly,

$$\begin{aligned} \Delta(T_{s,h}) &\leq 65, \\ N_{s,h} &= \left(2s + \frac{1}{63}\right) 64^h - \frac{1}{63}, \end{aligned} \tag{1}$$

$$\alpha_{s,h} = s64^h + \sum_{j=0}^{\lfloor (h-1)/2 \rfloor} 64^{h-1-2j}. \tag{2}$$

An empty sum is zero. Both root-conditioned independence numbers of the initial path are s ; the usual two-state maximum recurrence proves (2) by induction.

Theorem 1. *For every integer $s \geq 1$ there are $r_s \in (s - 1/64, s)$ and constants $c_s > 0$, $H_s < \infty$ with the following property. Put*

$$\rho_s = \frac{64r_s}{1 + r_s}, \quad \kappa_s = \log_{64} \rho_s.$$

For every $h \geq H_s$, $T_{s,h}$ has a consecutive interval $I_{s,h}$ such that

$$|I_{s,h}| \geq c_s \rho_s^h, \quad (3)$$

$$0 < k < \left(1 - \frac{1}{260(s+1)}\right) \alpha_{s,h} \quad (k \in I_{s,h}), \quad (4)$$

$$i_k(T_{s,h})^2 < i_{k-1}(T_{s,h}) i_{k+1}(T_{s,h}) \quad (k \in I_{s,h}).$$

In particular $\kappa_s \rightarrow 1$ as $s \rightarrow \infty$.

Corollary 2.

$$\boxed{\lim_{N \rightarrow \infty} \frac{\log M_{65}(N)}{\log N} = 1.}$$

Equivalently, $M_{65}(N) = N^{1-o(1)}$. More explicitly, for every $0 < \varepsilon < 1$ there are $c_\varepsilon > 0, N_\varepsilon$ such that $M_{65}(N) \geq c_\varepsilon N^{1-\varepsilon}$ for all $N \geq N_\varepsilon$.

The fixed degree is essential to the statement: it is 65 independently of ε . The distance from the top in (4), the run constant and the starting height may depend on ε . Neither a uniform bulk fraction as $\varepsilon \downarrow 0$ nor an $\Omega(N)$ run is asserted.

2 Tuning root odds with even paths

The independence polynomial of an n -vertex path is

$$Z_n(\lambda) = \sum_{k=0}^{\lceil n/2 \rceil} \binom{n-k+1}{k} \lambda^k.$$

The occupied/unoccupied odds at the root of $T_{s,0}$ are

$$R_s(\lambda) = \frac{\lambda Z_{2s-2}(\lambda)}{Z_{2s-1}(\lambda)}.$$

We seek a solution of

$$\lambda = r(1+r)^{64}, \quad R_s(\lambda) = r. \quad (5)$$

At this activity the odds are exactly r at every height, since the next odds are $\lambda/(1+r)^{64}$.

Lemma 3. Equation (5) has a solution $s - 1/64 < r < s$. Every such solution satisfies

$$2^{62} s^3 \leq \lambda \leq s(s+1)^{64}, \quad q := \frac{r}{1+r} \geq \frac{63}{127} > \frac{49}{100}, \quad 31 < \rho := 64q < 64.$$

Proof. Normalize by λ^s . The denominator coefficients, indexed by deficiency j , are $a_j = \binom{s+j}{2j}$; the numerator coefficients are $b_j = \binom{s+j}{2j+1}$. Their ratio is $b_j/a_j = (s-j)/(2j+1) \leq s$, strictly smaller for $j \geq 1$. Thus $R_s(\lambda) < s$. If $x = 4s^2/\lambda < 1$, then $a_j \leq (2s)^{2j}$, so the normalized denominator is at most $(1-x)^{-1}$. The numerator's constant coefficient is s . Consequently

$$R_s(\lambda) \geq s(1-x) = s - 4s^3/\lambda.$$

For $r_0 = s - 1/64$,

$$\lambda(r_0) \geq (63/64)^{65} s(s+1)^{64} > \frac{1}{4} s(s+1)^{64} \geq 2^{62} s^{33} \geq 2^{62} s^3.$$

Here $(63/64)^{65} > 1/4$ and $(s+1)^2 \geq 4s$ suffice. It follows that $R_s(\lambda(r_0)) \geq s - 2^{-60} > r_0$, whereas $R_s(\lambda(s)) < s$. Continuity gives a root between them. $\square$

Fix any such root for each s , and suppress s in the notation until the final counting step. Let X_h be the size of an independent set drawn with probability proportional to $\lambda^{|I|}$. Let $\mu_h^a = \mathbb{E}(X_h \mid \text{root} = a)$ for $a = 0, 1$, and $D_h = \mu_h^1 - \mu_h^0$.

Both initial conditional maxima equal s . Let $J_a = s - X_0$ conditional on the root state a . The normalized deficiency coefficients in both states are bounded by x^j , since $b_j/s \leq a_j$. Therefore

$$\mathbb{E}J_a \leq \frac{x}{(1-x)^2}, \quad \mathbb{E}J_a^2 \leq \frac{x(1+x)}{(1-x)^3}, \quad x \leq 2^{-60}/s.$$

In particular the convenient weaker bounds

$$|D_0| \leq 2^{-57}/s, \quad \text{Var}(X_0 \mid a) \leq 2^{-56}/s \quad (6)$$

hold. Root conditioning gives $D_{h+1} = 1 - \rho D_h$, hence

$$D_h = \theta + a_h, \quad \theta = \frac{1}{1+\rho}, \quad a_h = (D_0 - \theta)(-\rho)^h. \quad (7)$$

The coefficient a_0 is negative, $|a_0| > 1/128$, and $\theta/|a_0| < 2$.

3 The smoothing limit and an explicit approximation rate

Define centered conditional laws $Y_h^a = (X_h - \mu_h^a)/a_h$ and $z_h = D_h/a_h$. With independent child copies and $S_i \sim \text{Bernoulli}(q)$,

$$\begin{aligned} Y_{h+1}^0 &\stackrel{d}{=} -\rho^{-1} \sum_{i=1}^{64} (Y_{h,i}^{S_i} + (S_i - q)z_h), \\ Y_{h+1}^1 &\stackrel{d}{=} -\rho^{-1} \sum_{i=1}^{64} Y_{h,i}^0. \end{aligned} \quad (8)$$

At $z = 1$ this map contracts the maximum of the two centered-law W_2 distances by $\ell = 8/\rho < 1/3$: use independent centered couplings and add their second moments. Completeness gives a unique fixed pair (Y^0, Y^1) . Write

$$t = 64/\rho^2, \quad v_a = \text{Var } Y^a.$$

The fixed variance equations give

$$v_1 = tv_0, \quad v_0 = \frac{tq(1-q)}{1 - t(1-q) - qt^2}, \quad \frac{1-q}{4096} \leq v_a \leq \frac{5}{256}. \quad (9)$$

Indeed $t \leq 25/384$, the denominator is greater than $5/6$, and $t \geq 1/64$. With $p = 1 - q \leq 51/100$, we also have the useful bound

$$v_0 \leq \frac{5p}{128}, \quad v_1 = tv_0. \quad (10)$$

The initial W_2 distance is below $2^{-21} + 1/7 < 1/6$ by (6). The perturbation $z_h - 1 = \theta/a_h$ contributes at most $\ell\rho^{-h}$ at the next step. Summing the geometric series with ratio $1/8$ yields

$$\max_a W_2(Y_h^a, Y^a) < 2 \cdot 3^{-h}, \quad W_2(Z_h, Z) < 3 \cdot 3^{-h}, \quad (11)$$

where

$$m_h = \mathbb{E}X_h, \quad Z_h = (X_h - m_h)/a_h, \quad Z = Y^S + (S - q), \quad S \sim \text{Bernoulli}(q).$$

For the second bound use $Z_h = Y_h^S + (S - q)z_h$ and the same coupling.

Both fixed conditional laws are subgaussian with common proxy

$$\sigma^2 = \frac{t}{4(1-t)} < \frac{1}{50}. \quad (12)$$

To see this, a common proxy u is mapped to $t(u + 1/4)$ by (8), by the centered Bernoulli bound. Iterate from point masses and pass to the fixed point by truncation and Fatou. In particular $\mathbb{E}|Y^a| < 1/7$ and $\mathbb{E}|Z| < 3/5$.

4 A local limit theorem through two derivatives

This section supplies the finite-height step; convergence of moments alone would not suffice. Let $\psi_a(u) = \mathbb{E}e^{iuY^a}$ and $\phi(u) = \mathbb{E}e^{iuZ}$. Then

$$\psi_0(u) = \phi(-u/\rho)^{64}, \quad \psi_1(u) = \psi_0(-u/\rho)^{64}, \quad \phi(u) = pe^{-iqu}\psi_0(u) + qe^{ipu}\psi_1(u). \quad (13)$$

Put $Q(u) = \max_a |\psi_a(u)|$; thus $Q(u) \leq Q(u/\rho)^{64}$.

Lemma 4. *Set*

$$\tau = p/64, \quad \eta = \tau/\rho, \quad c = p^3/2^{38}, \quad \gamma = \log_\rho 64 > 1.$$

For $|u| \geq \eta$,

$$Q(u) \leq \exp[-c(|u|/\tau)^\gamma]. \quad (14)$$

The law of Z has a strictly positive entire density f .

Proof. For an independent copy Y' of Y^a , put $W = Y^a - Y'$. The subgaussian tail bound gives $\mathbb{E}W^4 \leq 64\sigma^4 < 1$. Since $\mathbb{E}W^2 = 2v_a$, the global inequality $\cos y \leq 1 - y^2/2 + y^4/24$ shows, for $|u| \leq \tau$,

$$|\psi_a(u)|^2 \leq 1 - v_a u^2 + u^4/24 \leq 1 - \frac{1}{2}v_a u^2.$$

Here $\tau^2 \leq p/4096 \leq v_a$. Hence $Q(u) \leq \exp(-pu^2/2^{14})$. On $[\eta, \tau]$ this is at most e^{-c} , since $\eta > p/4096$. For arbitrary $|u| \geq \eta$, choose $n \geq 0$ with $\eta \leq |u|/\rho^n \leq \tau$ and iterate the modulus recurrence. As $64^n \geq (|u|/\tau)^\gamma$, this proves (14).

The decay, with $\gamma > 1$, makes the inverse Fourier transform of ϕ entire: all exponential factors $e^{A|u|}$ are integrable against its modulus. Its real restriction is the nonnegative probability density f . This analytic function is nonzero, so its real zeros are isolated and $f > 0$ almost everywhere on $\mathbb{R}$. The first equation in (13) expresses the density of Y^0 as a scaled and reflected 64-fold convolution of f . Such a convolution is strictly positive everywhere (already the convolution of two almost-everywhere-positive densities has this property). The mixture formula then makes f itself strictly positive everywhere. $\square$

Let ϕ_h be the characteristic function of Z_h and define the exact Fourier interpolant

$$g_h(x) = \frac{1}{2\pi} \int_{-\pi|a_h|}^{\pi|a_h|} e^{-iux} \phi_h(u) du. \quad (15)$$

At every lattice node $x_{h,k} = (k - m_h)/a_h$, Fourier inversion gives

$$g_h(x_{h,k}) = |a_h| \mathbb{P}(X_h = k). \quad (16)$$

Lemma 5. *For each fixed s , $g_h \rightarrow f$ uniformly on $\mathbb{R}$ together with its first and second derivatives.*

Proof. Use physical characteristic functions $Q_h(v) = \max_a |\mathbb{E}(e^{ivX_h} | a)|$. Root conditioning gives $Q_h(v) \leq Q_k(v)^{64^{h-k}}$ for $0 \leq k \leq h$. Choose explicitly

$$h_0 = \max\{1, \lceil \log_3(8\tau/c) \rceil\}.$$

By (11), for $k \geq h_0$ and $\eta \leq |u| \leq \tau$,

$$Q_k(u/a_k) \leq e^{-c} + 2\tau 3^{-k} \leq e^{-c} + c/4 \leq e^{-c/4}.$$

The final inequality follows from $c < 1$ and $e^{-c} \leq 1 - c/2$. If $|u| \geq \eta$ and $|u/a_h| \leq \eta/|a_{h_0}|$, select $k \in [h_0, h]$ so that $\eta \leq |a_k u/a_h| \leq \tau$. The same shell argument gives

$$|\phi_h(u)| \leq \exp[-(c/4)(|u|/\tau)^\gamma]. \quad (17)$$

For completeness, the complementary physical frequencies have a separate lattice estimate. At height one, both conditional laws give positive mass to two adjacent sizes: use the empty set and a singleton when the root is absent, and the root alone and the root plus a vertex at distance two when it is present. The latter vertex exists also for $s = 1$. Each of these probabilities is at least $(1 + \lambda)^{-N_{s,1}}$, since $\lambda \geq 1$. For $|v| \leq \pi$, the elementary adjacent-atom bound therefore gives

$$Q_1(v) \leq e^{-d_s v^2}, \quad d_s = \frac{1}{8\{1 + s(s+1)^{64}\}^{256s+2}} > 0.$$

Indeed $|\chi(v)|^2 \leq 1 - 2ab(1 - \cos v)$ for adjacent masses a, b , and $1 - \cos v \geq 2v^2/\pi^2$. For $|u/a_h| \geq \eta/|a_{h0}|$ this implies

$$|\phi_h(u)| \leq \exp \left[-d_s \frac{\eta^2}{|a_{h0}|^2} 64^{h-1} \right] \quad (|u| \leq \pi|a_h|). \quad (18)$$

Its integral against $|u|^j$, $j \leq 2$, tends to zero, since the integration range grows only exponentially in h , whereas the negative exponent grows as 64^h . On the low part, (17) is a common integrable majorant after multiplication by $|u|^j$. Pointwise convergence follows from (11). Splitting at a fixed frequency, then letting that frequency tend to infinity, proves the uniform convergence of the three inverse transforms. This also accounts for the growing Fourier cutoff in (15). $\square$

5 A valley that survives as the root odds diverge

The small component has weight $p \rightarrow 0$ when $s \rightarrow \infty$. A variance bound independent of p would not establish a persistent valley. Bound (10) supplies the needed dependence.

Consider three intervals of radius $1/4$, with centers $-q$, $1/2 - q$, $1 - q$, and let their Z -probabilities be L, V, R . Chebyshev's inequality and (10) give

$$\begin{aligned} L &\geq p(1 - 16v_0) > \frac{2}{3}p, \\ R &\geq q(1 - 16v_1) > \frac{2}{3}p, \\ V &\leq 16(pv_0 + qv_1) \leq \frac{5}{8}p(p + t) < \frac{3}{8}p. \end{aligned} \quad (19)$$

For the numerical inequalities use $p \leq 51/100$, $q \geq 49/100$ and $t \leq 25/384$: $1 - (5/8)(51/100) > 2/3$, $(49/100)[1 - (5/8)(25/384)(51/100)] > (2/3)(51/100)$, and $(5/8)(51/100 + 25/384) < 3/8$.

If f were log-concave on the union of these three intervals, extend its restriction by zero outside that interval. Integrating a log-concave function over translates of a fixed interval preserves log-concavity (the one-dimensional Prékopa–Leindler inequality). This would give $V^2 \geq LR$, contradicting (19). Thus f is not log-concave on this interval. Its strict positivity and smoothness imply that $(\log f)'' > 0$ at some interior point. There are a closed interval $J_s \subset [-5/4, 5/4]$ of positive length and $\zeta_s > 0$ such that

$$\inf_{J_s} f > 0, \quad (\log f)'' \geq 2\zeta_s \quad \text{on } J_s. \quad (20)$$

Lemma 5 now implies $g_h > 0$ and $(\log g_h)'' \geq \zeta_s$ throughout J_s for every sufficiently large h .

Take all integers k whose three neighboring nodes lie in J_s . They form a consecutive interval, with at least $|J_s||a_h| - 3$ elements. Twice integrating the curvature inequality with the triangular kernel gives

$$\log \frac{g_h(x_{h,k-1})g_h(x_{h,k+1})}{g_h(x_{h,k})^2} \geq \zeta_s/|a_h|^2 > 0.$$

The Gibbs tilt cancels exactly:

$$\frac{\mathbb{P}(X_h = k-1)\mathbb{P}(X_h = k+1)}{\mathbb{P}(X_h = k)^2} = \frac{i_{k-1}(T_{s,h})i_{k+1}(T_{s,h})}{i_k(T_{s,h})^2}. \quad (21)$$

This proves (3) and all the strict inequalities on that run.

Bulk location and the extremal exponent. Set $\pi_* = q/(1+q)$ and $p_j = \pi_* + (q - \pi_*)(-q)^j$. The root state at distance j from the top has occupation probability p_j . Consequently the exact mean and its limit are

$$m_h = 64^h(\mu_0^0 + p_h D_0) + \sum_{j=0}^{h-1} 64^j p_j, \quad (22)$$

$$\frac{m_h}{64^h} \longrightarrow L_s := \mu_0^0 + \pi_* D_0 + \frac{\pi_*}{63}, \quad \frac{\alpha_{s,h}}{64^h} \longrightarrow A_s := s + \frac{64}{4095}.$$

The seed mean mixture is at most s , and $\pi_* < 1/2$, so

$$A_s - L_s > \frac{64}{4095} - \frac{1}{126} = \frac{1}{130}.$$

Also $L_s > 0$ by the deficiency bounds. Since $a_h/64^h = a_0(-q)^h \rightarrow 0$ and J_s is bounded, all its nodes satisfy (4) at every sufficiently large height. This proves Theorem 1.

Finally $r_s \in (s - 1/64, s)$ gives

$$\kappa_s = 1 - \frac{\log(1 + 1/r_s)}{\log 64} \longrightarrow 1.$$

For $0 < \varepsilon < 1$, $s = 1 + \lceil 1/\varepsilon \rceil$ already ensures $\kappa_s > 1 - \varepsilon$. By (1), the run is $\Omega_s(N_{s,h}^{\kappa_s})$. Consecutive tree orders satisfy $N_{s,h+1} = 64N_{s,h} + 1$, so maximizing over trees of order at most N extends the bound to every sufficiently large N , at the cost of a constant factor. The upper bound $M_{65}(N) \leq N$ proves Corollary 2. $\square$

6 A fully specified finite-height certificate: the seed P_4

We now quantify one instance without appealing to the qualitative selection of J_s in (20). For $s = 2$, let $r = 2 - \delta$, where

$$\lambda = (2 - \delta)(3 - \delta)^{64}, \quad \delta = \frac{5\lambda + 2}{\lambda^2 + 3\lambda + 1}, \quad 0 < \delta < 1/64. \quad (23)$$

This is (5) for P_4 . The conditional initial means are

$$\mu_0^0 = \frac{3\lambda + 2\lambda^2}{1 + 3\lambda + \lambda^2}, \quad \mu_0^1 = 1 + \frac{2\lambda}{1 + 2\lambda}.$$

Define q, ρ, D_0, a_h, m_h as above, and set

$$x_* = \frac{7}{40}, \quad d = 2^{-200}, \quad J = [x_* - d, x_* + d], \quad I_h = \{k \in \mathbb{Z} : x_{h,k-1}, x_{h,k}, x_{h,k+1} \in J\}. \quad (24)$$

Theorem 6. *For every integer $h \geq 512$, the tree $T_{2,h}$ satisfies*

$$|I_h| \geq 2^{-206} \rho^h \geq 2^{-209} N_{2,h}^{9/10}, \quad (25)$$

$$0 < k < \frac{997}{1000} \alpha_{2,h} \quad (k \in I_h), \quad (26)$$

$$\log \frac{i_{k-1}(T_{2,h}) i_{k+1}(T_{2,h})}{i_k(T_{2,h})^2} \geq \frac{2048}{|a_h|^2} > 0 \quad (k \in I_h). \quad (27)$$

6.1 Validated limiting characteristic functions

All interval computations below are implemented in `verify_fixed_degree.py`, using Python-FLINT 0.9.0 at 512-bit precision. The algebraic enclosure starts with $[0, 1/64]$ for δ and applies the right side of (23) twelve times using ball arithmetic. Each image continues to contain the root by its fixed-point identity; root existence was proved in Lemma 3. The computation gives

$$85/2 < \rho < 43, \quad 0 < D_0 < 2^{-99}, \quad 2^{-6} < |a_0| < 2^{-5}, \quad \theta/|a_0| < 2. \quad (28)$$

The normalized initial variances are below 2^{-42} , so the same approximation bounds (11) apply.

The exponential-moment coefficients $A_k = \mathbb{E}(Y^0)^k/k!$ and $C_k = \mathbb{E}(Y^1)^k/k!$ for $0 \leq k \leq 31$ are obtained by exact triangular equations with interval coefficients. Set $A_0 = C_0 = 1$, $A_1 = C_1 = 0$. When solving at order k , first set the two unknown coefficients to zero and call R_k, S_k the coefficients of z^k in

$$\{pe^{-qz}A(z) + qe^{pz}C(z)\}^{64}, \quad A(z)^{64}.$$

Writing $u_k = (-1/\rho)^k$ and $w_k = 64u_k$, the solution is

$$A_k = \frac{u_k R_k + w_k q u_k S_k}{1 - w_k p - w_k^2 q}, \quad C_k = w_k A_k + u_k S_k.$$

The verifier bounds all these denominators below by $9/10$.

From (12), $\mathbb{E}|Y^a|^{32}/32! \leq 2e/10^{32} < 6/10^{32}$. To approximate $\psi_a(t)$ for $|t| \leq 2^{19}$, evaluate the degree-31 Taylor polynomials at iu , $u = t/(-\rho)^5$, and apply five exact transformations (13). Since $2^{19}/42^5 < 1/200$, the initial error is at most $\epsilon = 6/2000^{32}$. Convex combinations and powers propagate it to at most $(1 + \epsilon)^{64^5} - 1 \leq 2 \cdot 64^5 \epsilon$. Thus the entire approximants P_a and their mixture P satisfy

$$|P_a(t) - \psi_a(t)| \leq E := 2^{-300}, \quad |P(t) - \phi(t)| \leq E \quad (|t| \leq 2^{19}). \quad (29)$$

Algebraic and moment coefficient uncertainty is included in each ball evaluation.

6.2 Whole-annulus coverage and finite-height tails

Put $T = 8192 = 2^{13}$ and $B = 524288 = 2^{19}$. Divide $[T, B]$ into 64512 intervals of width 8. At the midpoint t_c of each interval put $u_c = -t_c/\rho$. The full interval is covered by the bounds

$$\left(|P(u_c)| + E + \frac{3}{5} \frac{4}{42}\right)^{64}, \quad \left(|P_0(u_c)| + E + \frac{1}{7} \frac{4}{42}\right)^{64}. \quad (30)$$

These use the first absolute moments from Section 3 as Lipschitz constants before the last 64th power. Every cell bound is less than 2^{-240} ; the largest is below $3.363 \cdot 10^{-75}$. This certifies the complete continuum annulus, not just its midpoints.

For every $k \geq 192$, (11) transfers the annulus to height k , because

$$2^{-240} + 2B3^{-192} < U := 2^{-238}.$$

The physical recursion used in Lemma 5 shows that on each shell $T\rho^n \leq |t| \leq T\rho^{n+1}$ in the low physical-frequency part, the finite-height modulus is at most U^{64^n} . The same bound holds for the limit. For $0 \leq j \leq 3$ the resulting tail integrals (with the factor $1/\pi$ included) are at most

$$B_j^{\text{tail}} = 2^{19(j+1)+1-238}. \quad (31)$$

Indeed bound each shell endpoint by $B64^n$ and sum the geometric majorant: the ratio is at most $64^{j+1}U^{63} < 1/2$.

For the remaining physical frequencies one can use the seed itself. In root state zero the adjacent top masses have product $3\lambda^3/(1+3\lambda+\lambda^2)^2$; in state one their product is $2\lambda/(1+2\lambda)^2$.

The verifier bounds both below by 2^{-108} . Thus $Q_0(v) \leq \exp(-2^{-112}v^2)$ for $|v| \leq \pi$. Since $|a_{192}| < 64^{192}$, the high-frequency range starts above $T/|a_{192}| > 2^{-1139}$ in physical frequency, and its modulus is at most

$$\exp[-2^{6h-2390}].$$

Its Fourier integrals against $|t|^j/\pi$, $j \leq 3$, are below 2^{-500} for every $h \geq 512$: it suffices that $2^{6h-2390} > 24h + 508$, checked at 512 and preserved thereafter. The integration range is at most $\pi|a_h| < 2^{6h+2}$.

6.3 Validated curvature and transfer to every height

The code performs Acb validated integration on the 256 intervals of length 32 partitioning $[0, T]$, for each derivative order $j = 0, 1, 2$:

$$\frac{1}{\pi} \operatorname{Re} \int_0^T (-it)^j P(t) e^{-ix_* t} dt.$$

Absolute and relative tolerances are 2^{-180} . The integrands are entire. Adding the error $ET^{j+1} + B_j^{\text{tail}}$ certifies

$$\begin{aligned} 2.4 \cdot 10^{-18} &< f(x_*) < 2.6 \cdot 10^{-18}, \\ 8.3 \cdot 10^{-16} &< f'(x_*) < 8.5 \cdot 10^{-16}, \\ 5.0 \cdot 10^{-13} &< f''(x_*) < 5.2 \cdot 10^{-13}. \end{aligned} \tag{32}$$

These decimal endpoints are exact rational numbers. In particular, for $W = ff'' - (f')^2$, $W(x_*) > 2^{-101}$.

The central Fourier interval and (31) also give the global bounds $\|f^{(j)}\|_\infty < 2^{13(j+1)}$ for $0 \leq j \leq 3$. Thus $\|W'\|_\infty < 2^{66}$. On J this proves

$$W > 2^{-102}, \quad 2^{-59} < f < 2^{-58}, \quad |f'| < 2^{-50}, \quad |f''| < 2^{-40}. \tag{33}$$

Combining (11), the two Fourier tails and the high-frequency estimate gives, uniformly for every $h \geq 512$,

$$\|g_h - f\|_\infty < 2^{-216}, \quad \|g'_h - f'\|_\infty < 2^{-197}, \quad \|g''_h - f''\|_\infty < 2^{-178}. \tag{34}$$

For derivative order j , a sufficient bound on the error is $3 \cdot 3^{-h} T^{j+2} + 2B_j^{\text{tail}} + 2^{-500}$; the verifier compares this with the three displayed constants at $h = 512$.

Write these constants as R_0, R_1, R_2 . On J , we obtain $2^{-60} < g_h < 2^{-57}$, $|g''_h| < 2^{-39}$, and $|g'_h| + |f'| < 2^{-48}$. Hence

$$|g_h g''_h - (g'_h)^2 - W| \leq R_0 2^{-39} + 2^{-58} R_2 + 2^{-48} R_1,$$

which is small enough to give $g_h g''_h - (g'_h)^2 > 2^{-103}$. Consequently $(\log g_h)'' \geq 2^{-103}/2^{-114} = 2048$ on J . Equations (16) and (21) now prove (27).

The node count is at least $2d|a_h| - 3 \geq d|a_h| \geq 2^{-206} \rho^h$, since $2^{-206} 42^{512} \geq 4$. Furthermore, $(85/2)^{10} > 64^9$ and $N_{2,h} < 5 \cdot 64^h$ yield (25). To check the location directly, note $\mu_0^0 > 1$, $D_0 > 0$, $\pi_* < 2/5$ and $q < 2/3$. For $|x| < 1$, formula (22) gives

$$0 < \frac{m_h + a_h x}{64^h} \leq 2 + \frac{2}{315} + 2(2/3)^h < \frac{997}{1000} \left(2 + \frac{64}{4095} - 64^{-h} \right) \leq \frac{997}{1000} \frac{\alpha_{2,h}}{64^h}.$$

The strict middle inequality is checked at 512 and improves with h . All nodes used in (24) have $|x| < 1$. This completes the proof of Theorem 6. $\square$

7 Reproduction and relation to prior work

Run

```
python -m pip install python-flint==0.9.0
python verify_fixed_degree.py
```

The verifier writes `fixed_degree_certificate.json`, including its own SHA-256 hash, all rational comparisons, the annulus bound and the three certified derivative intervals. The executed run passed 855 checks, including all 64512 annulus cells and 768 validated integration segments. It does not construct the astronomically large trees or enumerate their coefficients; the recurrences and all-height arguments above are what transfer the finite computations to those trees.

Recent work already establishes arbitrarily many failures [1], several consecutive failures [2], and unbounded consecutive runs with an $\Omega(N^{1/3})$ bound in a reproducibility archive [3]. Arbitrarily many failures alone are therefore not the point of this note. Its specific assertion is the near-linear extremal exponent with one fixed degree bound, obtained by tuning the initial path while keeping branching 64. This comparison describes the sources inspected; it is not an exhaustive priority determination.

References

- [1] C. Bautista-Ramos, *Multiple breaks of log-concavity in the independence polynomials of trees*, arXiv:2511.00334 (2025). <https://arxiv.org/abs/2511.00334>.
- [2] C. Bautista-Ramos, C. Guillén-Galván, P. Gómez-Salgado, *Linear recurrences for non-log-concave independence polynomials of trees*, arXiv:2603.14204 (2026); *Graphs and Combinatorics* 42, 59. <https://doi.org/10.1007/s00373-026-03054-4>.
- [3] X. Yu, *Unbounded runs of non-log-concavity in independence polynomials of trees: reproducibility archive*, version 1.1.0 (3 September 2026). <https://doi.org/10.5281/zenodo.22263331>.